MATH-101 Calculus and Analytical Geometry – 3-0 CHs

By

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EQUATIONS OF LINES AND PLANES

Lec # 5

OBJECTIVE

- In this lecture we will discuss parametric equations of lines in 3-space. In 3-space, parametric equations of lines are especially important because they generally provide the most convenient form for representing lines algebraically.

- In this lecture we will use vectors to derive equations of planes in 3-space, and then we will use these equations to solve various geometric problems.

Note!!

To write the equation of a line in 3D space, we need a point on the line and a parallel vector to the line.

EQUATIONS OF LINES AND PLANES

Lines in 3D Space

Consider the line L through the point $P(x_0, y_0, z_0)$ that is parallel to the vector $\vec{v} = (a, b, c)$.

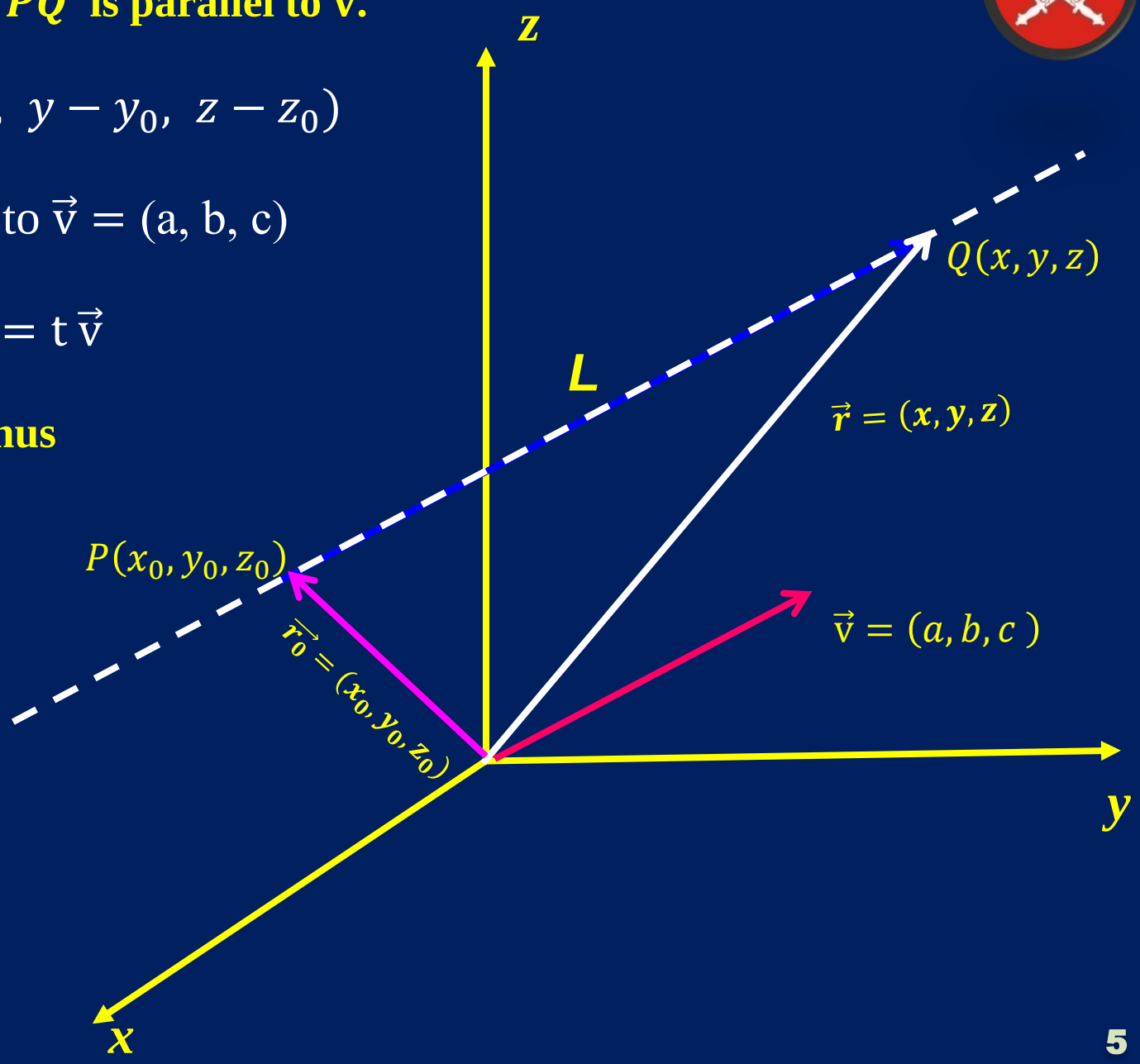
The line *L* consists of all points Q = (x, y, z) for which the vector $\overrightarrow{PQ}$ is parallel to $\vec{v}$.

Now, $\overrightarrow{PQ} = (x - x_0,\ y - y_0,\ z - z_0)$

Now, $\overrightarrow{PQ}$ is parallel to $\vec{v} = (a, b, c)$

$$\overrightarrow{PQ} = t\,\vec{v}$$

where t is a scalar. Thus



$$(x - x_0,\ y - y_0,\ z - z_0) = \overrightarrow{PQ} = t\vec{\mathrm{v}} = (ta,\ tb,\ tc)$$

Rewriting this equation gives

$$(x, y, z) - (x_0, y_0, z_0) = (ta, tb, tc)$$

Solving for the vector (x, y, z) gives

$$(x, y, z) = (x_0, y_0, z_0) + t(a, b, c)$$

Setting $\vec{\boldsymbol{r}} = (\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z})$, $\overrightarrow{r_0} = (x_0, y_0, z_0)$ and $\vec{\mathrm{v}} = (a,\ b,\ c)$

we get the following vector equation of a line

$$\vec{\boldsymbol{r}} = \overrightarrow{r_0} + t\vec{v}$$

Vector Equation Of a Line In 3D Space

The vector equation of a line in 3D space is given by the equation

$$\vec{r} = \overrightarrow{r_0} + t\vec{v}$$

Where $\overrightarrow{r_0} = (x_0, y_0, z_0)$ is a vector whose components are made of the point (x_0, y_0, z_0) on the line L and $\vec{v} = (a,\ b,\ c)$ are components of a vector that is parallel to the line L.

REMARK: If we take the vector equation

$$(x, y, z) = (x_0, y_0, z_0) + t(a, b, c)$$

and rewrite the right-hand side of this equation as one vector, we obtain

$$(x, y, z) = (x_0 + ta, y_0 + tb, z_0 + tc)$$

Equating components of this vector gives the **parametric equations of a line.**

$$x = x_0 + ta, \qquad y = y_0 + tb, \qquad z = z_0 + tc$$

Example: Find a vector equation for the line through $(4, 6, -3)$ and parallel to vector $5\hat{\imath} - 10\hat{\jmath} + 2\hat{k}$.

Solution :

$$\overrightarrow{r_0} = (x_0, y_0, z_0) = (4, 6, -3)$$

$$\vec{v} = (\mathrm{a}, \mathrm{b}, \mathrm{c}) = 5\hat{\imath} - 10\hat{\jmath} + 2\hat{k}$$

Using vector equation for the line

$$\vec{\boldsymbol{r}} = \overrightarrow{r_0} + t\vec{v}$$

$$(x, y, z) = (x_0, y_0, z_0) + t(a, b, c)$$

$$= (4, 6, -3) + t(5, -10, 2)$$

$$= (4 + 5t, 6 - 10t, -3 + 2t)$$

So $x = 4 + 5t, \quad y = 6 - 10t, \quad z = -3 + 2t$

Example: Find a vector equation for the line through $(2, -1, 8)$ and $(5, 6, -3)$.

Solution :

If we label the points as $P(2, -1, 8)$ and $Q(5, 6, -3)$, then a direction vector for the line through P and Q is

$$\vec{v} = \overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP} = (5 - 2, 6 + 1, -3 - 8) = (3, 7, -11)$$

From the vector equation of that line

$$\vec{r} = \overrightarrow{r_0} + t\vec{v}$$

$$(x, y, z) = (2, -1, 8) + t(3, 7, -11)$$

$$= (2 + 3t, -1 + 7t, 8 - 11t)$$

So $x = 2 + 3t, \quad y = -1 + 7t, \qquad z = 8 - 11t$

Example: Find a vector equation for the line through $(5, 2,4)$ and parallel to vector $4\hat{\imath} + 7\hat{\jmath} - 9\hat{k}$.

Solution :

$$\overrightarrow{r_0} = (x_0, y_0, z_0) = (5, 2, 4)$$

$$\vec{v} = (\mathrm{a}, \mathrm{b}, \mathrm{c}) = 4\hat{\imath} + 7\hat{\jmath} - 9\hat{k}$$

Using vector equation for the line

$$\vec{\boldsymbol{r}} = \overrightarrow{r_0} + t\vec{v}$$

$$(x, y, z) = (x_0, y_0, z_0) + t(a, b, c)$$

$$= (5, 2, 4) + t(4, 7, -9)$$

$$= (5 + 4t, 2 + 7t, 4 - 9t)$$

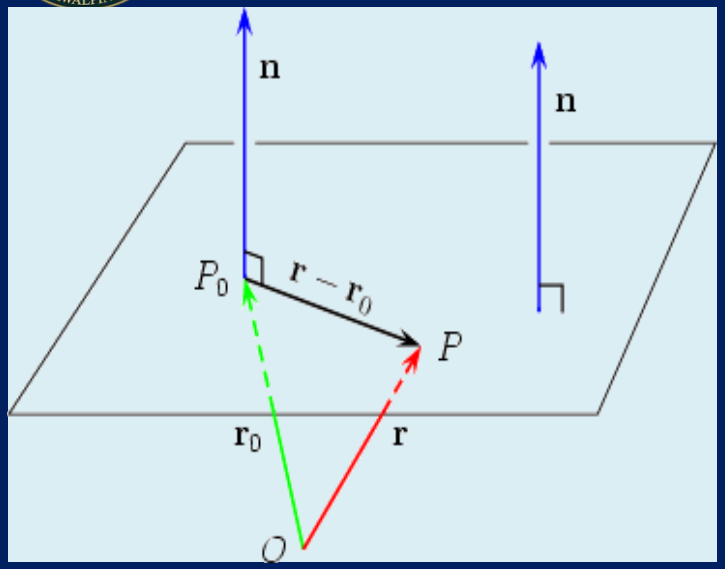
So $x = 5 + 4t, \quad y = 2 + 7t, \qquad z = 4 - 9t$

PLANES IN 3D SPACE

Consider the Plane containing the point $P(x_0, y_0, z_0)$ and normal vector $\vec{n} = (a, b, c)$ perpendicular to the plane.

Note!!

To write the equation of a plane in 3D space, we need a point on the plane and a vector normal (orthogonal) to the plane.



$\vec{n} = (a, b, c)$

z

$Q(x, y, z)$

$P(x_0, y_0, z_0)$

y

x

The plane consists of all points Q = (x, y, z) for which the vector $\overrightarrow{PQ}$ is orthogonal to the normal vector $\vec{n}$. Since $\overrightarrow{PQ}$ and $\vec{n}$ are orthogonal , the following equations hold:

Now, $$\vec{n} \cdot \overrightarrow{PQ} = 0$$

$$(a, b, c) \cdot (x - x_0, y - y_0, z - z_0) = 0$$

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

This gives the standard equation of a plane.

If we expand this equation, we obtain the following equation:

$$ax + by + cz - ax_0 - by_0 - cz_0 = 0$$

Setting $d = -ax_0 - by_0 - cz_0$ gives the general form of the equation of a plane in 3D space

$$ax + by + cz + d = 0$$

Example: Find an equation of the plane that contains the point $(4, -1, 3)$ and is perpendicular to the vector $\vec{n} = 2\hat{\imath} + 8\hat{\jmath} - 5\hat{k}$.

Solution: It follows immediately from the equation of the plane containing $P(x_0, y_0, z_0)$ and with normal vector $\vec{n} = a\hat{\imath} + b\hat{\jmath} + c\hat{k}$, that is,

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

That the identification $x_0 = 4, y_0 = -1, z_0 = 3, a = 2, b = 8, c = -5$ yields

$$2(x - 4) + 8(y + 1) - 5(z - 3) = 0$$

Example: Finding an equation for a plane through three points

Find an equation for the plane through $A(0,0,1)$, $B(2,0,0)$ and $C(0,3,0)$.

Solution: We find a vector normal to the plane and use it with one of the points (it does not matter which) to write an equation for the plane:

The Cross Product

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \mathrm{i} & \hat{J} & \hat{\mathrm{k}} \\ 2 & 0 & -1 \\ 0 & 3 & -1 \end{vmatrix} = 3\mathrm{i} + 2\hat{J} + 6\,\hat{\mathrm{k}}$$

is normal to the plane. We substitute the components of this vector and the coordinates of $A(0,0,1)$ into the component form of the equation to obtain $(a(x - x_0) + b(y - y_0) + c(z - z_0) = 0)$

$$3(x - 0) + 2(y - 0) + 6(z - 1) = 0$$

$$3x + 2y + 6z = 6$$

EXERCISE

Question # 1:

Find the vector equations for the line through the point $(1, 0, -3)$ and parallel to the vector $2\hat{\imath} - 4\hat{\jmath} + 5\hat{k}$.

Question # 2 :

Find the vector equations for the line through the points $(1, 2, 0)$ and $(-5, 4, 2)$

Question # 3 :

Find the equation of the plane through the point $(-4, 3, 1)$ that is perpendicular to the vector $-4\hat{\imath} + 7\hat{\jmath} - 2\hat{k}$

Question # 4 :

Find the equation of the plane through the point $(-3, 0, 7)$ that is perpendicular to the vector $5\hat{\imath} + 2\hat{\jmath} - \hat{k}$